

# **SIR C R REDDY COLLEGE OF ENGINEERING (AUTONOMOUS)**

Eluru-534007, West Godavari Dist, Andhra Pradesh, India

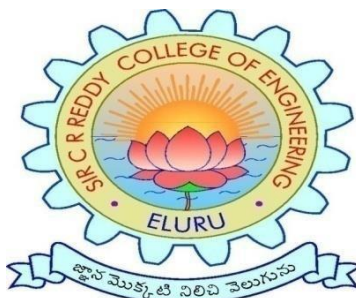
(Approved by AICTE, New Delhi & Permanently affiliated to JNTUK, Kakinada)

Accredited by NBA, NAAC with 'A' Grade

Telephone No: 08812-230840, 230565, Fax: 08812-224193

Website: [www.sircrrengg.ac.in](http://www.sircrrengg.ac.in)

## **COURSE HANDOUT**



## **DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**

### **SOFTWARE TECHNOLOGY AND METHDOLOGIES**

#### **III B.Tech IISemester**

#### **R23 Regulation**

**Academic Year: 2025-2026**

| Name of the Staff  | CSE A      | CSE B |  |  |
|--------------------|------------|-------|--|--|
|                    |            |       |  |  |
| Course Coordinator | K.PAVITHRA |       |  |  |

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SIR CR RIDDY COLLEGE OF ENGINEERING - ELURU

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TIME TABLE IN-CHARGE

HOD, CSE

Head of the Department  
Computer Science & Engineering  
Sir C. R. R. College of Engineering  
ELURU - 534 007

PRJ C. I. T. A. L

Sir C. R. R.

Principal  
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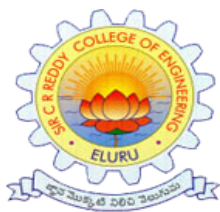
**DEPARTMENT OF ARTIFICIAL INTELLIGENCE & DATA SCIENCE  
DEPARTMENT OF CSE- (AI&ML)**

## **VISION OF THE DEPARTMENT**

To be a premier department in the region with an integrated teaching, learning and research environment that enables to address the challenges of the core and emerging technologies with social responsibility.

## **MISSION OF THE DEPARTMENT**

- M1:** Amplify the scholarliness of the students in theoretical and practical areas of the discipline.
- M2:** Application of knowledge gained to invent and develop new products.
- M3:** Refreshing the existing technologies to meet the dynamic needs and challenges of the Industry.
- M4:** Develop innovative technocrats with ethical and social responsibility.



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|                           |                                            |
|---------------------------|--------------------------------------------|
| <b>Course Title :</b>     | <b>SOFTWARE TECHNOGY AND METHODOLOGIES</b> |
| <b>Course Code:</b>       |                                            |
| <b>Year/Semester:</b>     | <b>3/4CSE</b>                              |
| <b>Class/Section</b>      | <b>A,B,C,D</b>                             |
| <b>A.Y</b>                | <b>2025-2026</b>                           |
| <b>Name of the Staff:</b> |                                            |

## **SYLLABUS**

### **UNIT - I**

Introduction: Purpose of testing, Dichotomies, model for testing, consequences of bugs, taxonomy of bugs Flow graphs and Path testing: Basics concepts of path testing, predicates, path predicates and achievable paths, path sensitizing, path instrumentation, application of path testing.

### **UNIT - II**

Transaction Flow Testing: transaction flows, transaction flow testing techniques.

Data Flow testing: Basics of data flow testing, strategies in data flow testing, application of data flow testing.

Domain Testing: domains and paths, Nice & ugly domains, domain testing, domains and interfaces testing, domain and interface testing, domains and testability.

### **UNIT - III**

Paths, Path products and Regular expressions: path products & path expression, reduction procedure, applications, regular expressions & flow anomaly detection.

Logic Based Testing: overview, decision tables, path expressions, kv charts, specifications.

### **UNIT - IV**

State, State Graphs and Transition testing: state graphs, good & bad state graphs, state testing, Testability tips.

## UNIT - V

Graph Matrices and Application: Motivational overview, matrix of graph, relations, power of a matrix, node reduction algorithm, building tools. (Student should be given an exposure to a tool like Jmeter/selenium/soapUI/Catalon).

### COURSE OBJECTIVES

- To provide knowledge of the concepts in software testing such as testing process, criteria, strategies, and methodologies.
- To develop skills in software test automation and management using the latest tools.

### COURSE OUTCOMES

At the end of the course students should be able to:

|            |                                                                                                                                                                                                                                |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CO1</b> | Understand the fundamental purpose of software testing, different testing models, bug taxonomy, and apply path testing techniques such as flow graphs, path predicates, and path sensitization to design effective test cases. |
| <b>CO2</b> | Apply transaction flow, data flow, and domain testing methods to detect logical and interface-related defects, and analyze the testability of software components.                                                             |
| <b>CO3</b> | Use path products, path expressions, and logic-based testing approaches (like decision tables and KV charts) to model and analyze program logic and detect flow anomalies.                                                     |
| <b>CO4</b> | Develop and interpret state graphs for software systems, identify valid and invalid transitions, and perform state-based testing to ensure correctness of software behavior.                                                   |
| <b>CO5</b> | Construct graph matrices, apply node reduction algorithms for software analysis, and gain hands-on experience using automated testing tools such as JMeter, Selenium, SoapUI, or Katalon for test execution and reporting.     |

### CO-PO/PSO MAPPING

| CO  | PO 1 | PO 2 | PO 3 | PO 4 | PO 5 | PO 6 | PO 7 | PO 8 | PO 9 | PO 10 | PO 11 | PO 12 | PSO 1 | PSO 2 |
|-----|------|------|------|------|------|------|------|------|------|-------|-------|-------|-------|-------|
| CO1 | -    | -    | -    | -    | -    | 2    | -    | 2    | -    | -     | -     | 2     | -     | -     |
| CO2 | -    | -    | -    | -    | -    | -    | -    | -    | 2    | -     | -     | 2     | -     | -     |
| CO3 | -    | -    | -    | -    | -    | 2    | 2    | 2    | -    | -     | -     | 2     | -     | -     |
| CO4 | -    | -    | -    | -    | -    | 2    | -    | 2    | -    | -     | -     | 2     | -     | -     |
| Avg | -    | -    | -    | -    | -    | 2.0  | 2.0  | 2.0  | 2.0  | -     | -     | 2.0   | -     | -     |

#### **Program outcomes:**

**PO1: Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

**PO2: Problem analysis:** Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using the first principles of mathematics, natural sciences, and engineering sciences.

**PO3: Design/development of solutions:** Design solutions for complex engineering problems and system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, society, and environmental considerations.

**PO4: Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

**PO5: Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

**PO6: The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.



**PO7: Environment and sustainability:** Understand the impact of the professional engineering solutions in society and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

**PO8: Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

**PO9: Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multi-disciplinary settings.

**PO10: Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

**PO11: Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multi-disciplinary environments.

**PO12: Life-long learning:** Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

### **Program Educational Objectives:**

**PEO1:** To produce graduates with strong foundation of domain knowledge in the field of Computer Science & Engineering.

**PEO2:** To produce graduates who can successfully pursue graduate studies and research in Computer Science & Engineering.

**PEO3:** To produce graduates who can practice their profession and communicate effectively either individually or in groups to meet dynamic needs of the industry.

**PEO4:** To imbibe in graduates a desire for lifelong learning moulded with professional and ethical values suitable to society.

### **Program Specific Outcomes:**

**PSO1:** Apply Data Science and Machine Learning concepts towards development of Innovative Applications.

**PSO2:** Apply standard practices and strategies in software project development using software tools to excel as a professional software developer.

**CO-PO MAPPING JUSTIFICATION**

| <b><u>CO</u></b> | <b><u>PO/PSO</u></b> | <b><u>CORELATION LEVEL</u></b> | <b><u>JUSTIFICATION</u></b>                                                                                                            |
|------------------|----------------------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| CO1              | PO6                  | 2                              | CO1 is moderately mapped with PO6 as to Understand the essentials of human values in society                                           |
|                  | PO8                  | 2                              | CO1 is moderately mapped with PO8 as to Understand the essentials of human values with professional ethics                             |
|                  | PO12                 | 2                              | CO1 is moderately mapped with PO12 for lifelong learning of human values                                                               |
| CO2              | PO8                  | 2                              | CO2 is moderately mapped with PO8 as to apply the harmony as an individual and as a member of family                                   |
|                  | PO12                 | 2                              | CO2 is moderately mapped with PO12 for lifelong to Apply the role of a human being for ensuring harmony in self and family             |
| CO3              | PO6                  | 2                              | CO3 is moderately mapped with PO6 as to Interpret the role of a human being in ensuring harmony in society                             |
|                  | PO7                  | 2                              | CO3 is moderately mapped with PO7 as to Interpret the role of a human being in ensuring harmony in society and nature                  |
|                  | PO8                  | 2                              | CO3 is moderately mapped with PO8 as to Interpret the role of a human being in ensuring ethical responsibilities in society and nature |
|                  | PO12                 | 2                              | CO3 is moderately mapped with PO12 as to Interpret the role of a human being in ensuring harmony for life long in society and nature   |
| CO4              | PO6                  | 2                              | CO4 is moderately mapped with PO6 as to Distinguish between ethical and unethical practices in society                                 |
|                  | PO8                  | 2                              | CO4 is moderately mapped with PO8 as to Distinguish between ethical and unethical practices in engineering practice                    |
|                  | PO12                 | 2                              | CO4 is moderately mapped with PO12 as to start working out the strategy to actualize a harmonious environment wherever they work       |

|                               | <b>CSE A</b> | <b>CSE B</b> | <b>CSE C</b> | <b>CSE D</b> |
|-------------------------------|--------------|--------------|--------------|--------------|
| <b>Name of the Staff</b>      | G.SRIDIVYA   | K.PAVITHRA   | B.RAMESH     | K.POOJITHA   |
| <b>Signature of the Staff</b> |              |              |              |              |
| <b>Course Coordinator</b>     | K.PAVITHRA   |              |              |              |

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## UNIVERSAL HUMAN VALUES 2: UNDERSTANDING HARMONY

### IV B.TECH I SEMESTER

**A.Y 2025-2026**

### **LESSON PLAN**

| SNo | No of Classes | Topics                                                                                                                                                                                                                                                                                        | Teaching Aids |
|-----|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| 1   | Class-1       | Introduction: Purpose of testing, Dichotomies, model for testing, consequences of bugs, taxonomy of bugs Flow graphs and Path testing: Basics concepts of path testing, predicates, path predicates and achievable paths, path sensitizing, path instrumentation, application of path testing | BB/PPT        |
| 2   | Class-2       | Dichotomies,                                                                                                                                                                                                                                                                                  | BB/PPT        |
| 3   | Class-3       | model for testing, consequences of bugs, taxonomy of Bugs                                                                                                                                                                                                                                     | BB/PPT        |
| 4   | Class-4       | taxonomy of bugs Flow graphs and Path testing:                                                                                                                                                                                                                                                | BB/PPT        |
| 5   | Class-5       | Basics concepts of path testing, predicates,                                                                                                                                                                                                                                                  | BB/PPT        |
| 6   | Class-6       | path predicates and achievable paths,                                                                                                                                                                                                                                                         | BB/PPT        |

|    |          |                                                                                                                                                                                                                                                                                                                                                                           |        |
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| 7  | Class-7  | path sensitizing                                                                                                                                                                                                                                                                                                                                                          | BB/PPT |
| 8  | Class-8  | path instrumentation                                                                                                                                                                                                                                                                                                                                                      | BB/PPT |
| 9  | Class-9  | path instrumentation                                                                                                                                                                                                                                                                                                                                                      | BB/PPT |
| 10 | Class-10 | application of path testing                                                                                                                                                                                                                                                                                                                                               | BB/PPT |
| 11 | Class-11 | application of path testing                                                                                                                                                                                                                                                                                                                                               | BB/PPT |
| 12 | Class-12 | TUTORIAL                                                                                                                                                                                                                                                                                                                                                                  | BB/PPT |
| 13 | Class-13 | Transaction Flow Testing: transaction flows, transaction flow testing techniques.<br>Data Flow testing: Basics of data flow testing, strategies in data flow testing, application of data flow testing.<br>Domain Testing: domains and paths, Nice & ugly domains, domain testing, domains and interfaces testing, domain and interface testing, domains and testability. | BB/PPT |
| 14 | Class-14 | transaction flows, transaction flow testing techniques.                                                                                                                                                                                                                                                                                                                   | BB/PPT |
| 15 | Class-15 | Data Flow testing: Basics of data flow testing,                                                                                                                                                                                                                                                                                                                           | BB/PPT |
| 16 | Class-16 | strategies in data flow testing                                                                                                                                                                                                                                                                                                                                           | BB/PPT |

|    |          |                                                                                                                                                                                                                                                                 |        |
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| 17 | Class-17 | application of data flow testing                                                                                                                                                                                                                                | BB/PPT |
| 18 | Class-18 | Domain Testing: domains and paths                                                                                                                                                                                                                               | BB/PPT |
| 19 | Class-19 | Nice & ugly domains,                                                                                                                                                                                                                                            | BB/PPT |
| 20 | Class-20 | domain testing                                                                                                                                                                                                                                                  | BB/PPT |
| 21 | Class-21 | domains and interfaces testing,                                                                                                                                                                                                                                 | BB/PPT |
| 22 | Class-22 | domains and testability                                                                                                                                                                                                                                         | BB/PPT |
| 23 | Class-23 | domains and testability                                                                                                                                                                                                                                         | BB/PPT |
| 24 | Class-24 | TUTORIAL                                                                                                                                                                                                                                                        | BB/PPT |
| 25 | Class-25 | Paths, Path products and Regular expressions: path products & path expression, reduction procedure, applications, regular expressions & flow anomaly detection.<br>Logic Based Testing: overview, decision tables, path expressions, kv charts, specifications. | BB/PPT |
| 26 | Class-26 | Path products and Regular expressions                                                                                                                                                                                                                           | BB/PPT |
| 27 | Class-27 | path products & path expression                                                                                                                                                                                                                                 | BB/PPT |
| 28 | Class-28 | reduction procedure,                                                                                                                                                                                                                                            | BB/PPT |
| 29 | Class-29 | applications, regular expressions                                                                                                                                                                                                                               | BB/PPT |

|    |          |                                                                                                                           |        |
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| 30 | Class-30 | flow anomaly detection.                                                                                                   | BB/PPT |
| 31 | Class-31 | Logic Based Testing: overview,                                                                                            | BB/PPT |
| 32 | Class-32 | decision tables                                                                                                           | BB/PPT |
| 33 | Class-33 | path expressions                                                                                                          | BB/PPT |
| 34 | Class-34 | kv charts,                                                                                                                | BB/PPT |
| 35 | Class-35 | specifications.                                                                                                           | BB/PPT |
|    |          |                                                                                                                           |        |
| 36 | Class-36 | <b>TUTORIAL</b>                                                                                                           | BB/PPT |
| 37 | Class-37 | State, State Graphs and Transition testing:<br>state graphs, good & bad state graphs, state<br>testing, Testability tips. | BB/PPT |
| 38 | Class-38 | State                                                                                                                     | BB/PPT |
| 39 | Class-39 | Graphs and Transition testing                                                                                             | BB/PPT |

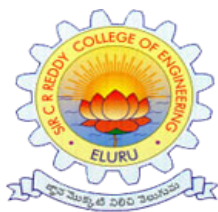
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| 40 | Class-40 | Graphs and Transition testing                                                                                                                                                                                                        | BB/PPT |
| 41 | Class-41 | state graphs,                                                                                                                                                                                                                        | BB/PPT |
| 42 | Class-42 | state graphs,                                                                                                                                                                                                                        | BB/PPT |
| 43 | Class-43 | good & bad state graphs                                                                                                                                                                                                              | BB/PPT |
| 44 | Class-44 | good & bad state graphs                                                                                                                                                                                                              | BB/PPT |
| 45 | Class-45 | state testing, Testability tips.                                                                                                                                                                                                     | BB/PPT |
| 46 | Class-46 | state testing, Testability tips.                                                                                                                                                                                                     | BB/PPT |
| 47 | Class-47 | state testing, Testability tips.                                                                                                                                                                                                     | BB/PPT |
| 48 | Class-48 | TUTORIAL                                                                                                                                                                                                                             | BB/PPT |
| 49 | Class-49 | Graph Matrices and Application: Motivational overview, matrix of graph, relations, power of a matrix, node reduction algorithm, building tools. (Student should be given an exposure to a tool like Jmeter/selenium/soapUI/Catalon). | BB/PPT |
| 50 | Class-50 | Motivational overview,                                                                                                                                                                                                               | BB/PPT |
| 51 | Class-51 | matrix of graph,                                                                                                                                                                                                                     | BB/PPT |

|    |          |                                                                                                     |        |
|----|----------|-----------------------------------------------------------------------------------------------------|--------|
| 52 | Class-52 | relations                                                                                           | BB/PPT |
| 53 | Class-53 | ,relations                                                                                          | BB/PPT |
| 54 | Class-54 | power of matrix                                                                                     | BB/PPT |
| 55 | Class-55 | power of matrix                                                                                     | BB/PPT |
| 56 | Class-56 | node reduction algorithm                                                                            | BB/PPT |
| 57 | Class-57 | node reduction algorithm                                                                            | BB/PPT |
| 58 | Class-58 | building tools. (Student should be given an exposure to a tool like Jmeter/selenium/soapUI/Catalon) | BB/PPT |
| 59 | Class-59 | building tools. (Student should be given an exposure to a tool like Jmeter/selenium/soapUI/Catalon) | BB/PPT |
| 60 | Class-60 | TUTORIAL                                                                                            | BB/PPT |

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| <b>Name of the Staff</b>      | <b>CSE</b> |  |  |  |
|                               |            |  |  |  |
| <b>Signature of the Staff</b> |            |  |  |  |
| <b>Course Coordinator</b>     |            |  |  |  |

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**DEPARTMENT OF ARTIFICIAL INTELLIGENCE & DATA SCIENCE  
DEPARTMENT OF CSE- (AI&ML)**

## **COURSE SCHEDULE**

### **UNIVERSAL HUMAN VALUES 2: UNDERSTANDING HARMONY**

**IV/IV B.Tech SEMESTER - I**

**A.Y: 2025-2026.**

**The schedule for the whole course/subject is:**

| <b>UNIT<br/>NO</b> | <b>DESCRIPTION OF<br/>THE CHAPTER</b> | <b>DESCRIPTION OF THE TOPICS</b>                                                                                                                                                                                                                                                                                                                                                 | <b>TOTAL<br/>NO. OF<br/>PERIODS</b> |
|--------------------|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| <b>1</b>           | <b>UNIT - I</b>                       | Introduction: Purpose of testing, Dichotomies, model for testing, consequences of bugs, taxonomy of bugs<br>Flow graphs and Path testing: Basics concepts of path testing, predicates, path predicates and achievable paths, path sensitizing, path instrumentation, application of path testing.                                                                                | <b>10+2</b>                         |
| <b>2</b>           | <b>UNIT - II</b>                      | Transaction Flow Testing: transaction flows, transaction flow testing techniques.<br>Data Flow testing: Basics of data flow testing, strategies in data flow testing, application of data flow testing.<br>Domain Testing: domains and paths, Nice & ugly domains, domain testing, domains<br><br>and interfaces testing, domain and interface testing, domains and testability. | <b>10+2</b>                         |

|   |                   |                                                                                                                                                                                                                                                              |      |
|---|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 3 | <b>UNIT - III</b> | Paths, Path products and Regular expressions: path products & path expression, reduction procedure, applications, regular expressions & flow anomaly detection. Logic Based Testing: overview, decision tables, path expressions, kv charts, specifications. | 12+2 |
| 4 | <b>UNIT - IV</b>  | State, State Graphs and Transition testing: state graphs, good & bad state graphs, state testing, Testability tips.                                                                                                                                          | 11+2 |
| 5 | <b>UNIT - V</b>   | Graph Matrices and Application: Motivational overview, matrix of graph, relations, power of a matrix, node reduction algorithm, building tools. (Student should be given an exposure to a tool like Jmeter/selenium/soapUI/Catalon).                         | 12+2 |

Total no of instructional periods available for the course : 65 periods

Total no of estimated periods : 60 periods

### Text Book

#### 1. Text Books:

1. Software Testing techniques - Baris Beizer, Dreamtech, second edition.

Software Testing Tools – Dr. K. V. K. K. Prasad, Dreamtech.

|                               |            |  |            |  |
|-------------------------------|------------|--|------------|--|
| <b>Name of the Staff</b>      | <b>CSE</b> |  | <b>CSE</b> |  |
|                               |            |  |            |  |
| <b>Signature of the Staff</b> |            |  |            |  |
| <b>Course Coordinator</b>     |            |  |            |  |

**HoD- CSE**